MIL-C-28613A(YD)
8 December 1976
SUPERSEDING
MIL-C-28613(YD)
30 April 1971

MILITARY SPECIFICATION

CRANES, OVERHEAD TRAVELING; ELECTRIC POWERED, 25-TON CAPACITY OR LESS

This specification is approved for use by the Naval Facilities Engineering Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers overhead traveling electric powered single and double girder cranes of the toprunning type, with capacities up to 25 tons. Overhead electric traveling (OET) cranes covered in this specification are not intended for heavy or severe duty use; for use in handling hot metal or fissionable material; or for use in hazardous areas as defined in National Fire Protection Association (NFPA) No. 70, article 500.

* 1.1.1 Definitions. To ensure clarity of understanding of the applicability of this specification, the below selected definitions of terms used herein are provided:

"Overhead Electric Traveling (OET) Crane". Any electrically powered crane with a movable bridge that travels on either the lower flanges ("underhung" or "underrunning" type) or the top surface of rails ("toprunning" type) of a fixed runway structure located above ground level.

"Toprunning Crane". A specific type of OET that runs on the upper flanges of rails of an overhead fixed runway structure and carrying a movable or fixed hoisting machanism.

* 1.2 Classification. Cranes shall be of the following types and service classes, as specified (see 6.2.1):

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer (156), Naval Construction Battalion Center, Port Hueneme, CA 93043, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 3950

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Type I - Single girder.
Type II - Double girder.

Class A1 - Standby service.
Class A2 - Infrequent service.
Class B - Light service.
Class C - Moderate service.
Class D - Heavy service.
Class E - Severe service.

(The service classes listed above roughly correspond to like-lettered classfications given in Crane Manufacturers Association of America (CMAA) No. 70 or 74. However, the design requirements specified herein differ in many respects from the design requirements of CMAA No. 70 or 74.)

* 2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

RR-W-410 - Wire Rope and Strand.

MILITARY

MIL-V-173 - Varnish, Moisture-and-Fungus Resistant (For Treatment of Communications, Electronic, and Associated Equipment).
MIL-P-514 - Plates, Identification, Instruction and Marking, Blank.

STANDARDS

MILITARY

MIL-STD-461 - Electromagnetic Interference Characteristics Requirements for Equipment.
MIL-STD-462 - Electromagnetic Interference Characteristics, Measurement of.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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* 2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

NATIONAL BUREAU OF STANDARDS (NBS)

H28 - Screw-Thread Standards for Federal Services.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402.)

AMERICAN GEAR MANUFACTURERS' ASSOCIATION (AGMA)

AGMA Standards as applicable.

(Application for copies should be addressed to the American Gear Manufacturers' Association, 1330 Massachusetts Avenue, N.W., Washington, DC 20005.)

AMERICAN INSITUTE OF STEEL CONSTRUCTION (AISC)

Design, Fabrication & Erection of Structural Steel of Buildings.

(Application for copies should be addressed to the American Institute of Steel Construction, Inc., 101 Park Avenue, New York, NY 10017.)

AMERICAN NATIONAL STANDARDS INSTITUTE, INC., (ANSI)

B30.2 - Safety Standard for Overhead and Gantry Cranes.
B30.16 - Safety Standards for Overhead Hoists.

(Application for copies should be addressed to the American National Standards Institute, Inc., 1430 Broadway, New York, NY 10018.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

A27 - Specifications for Mild- to Medium-Strength Carbon-Steel Castings for General Application.
A36 - Specification for Structural Steel.
A148 - Specification for High-Strength Steel Castings for Structural Purposes.
A235 - Specification for Carbon Steel Forgings for General Industrial Use.
A236 - Specification for Carbon Steel Forgings for Railway Use.

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A237 - Specification for Alloy Steel Forgings for General Industrial Use.
A439 - Specification for Austenitic Ductile Iron Castings.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

AMERICAN WELDING SOCIETY (AWS)

D1.1 - Structural Welding Code.

(Application for copies should be addressed to the American Welding Society, 2501 N.W., 7th Street, Miami, FL 33125.)

CRANE MANUFACTURERS ASSOCIATION OF AMERICA, INC. (CMAA)

No. 70 - Specification for Electric Overhead Traveling Cranes.
No. 74 - Specification for Toprunning and Underrunning Single Girder Electric Overhead Traveling Cranes.

(Application for copies should be addressed to the Crane Manufacturers Association of America, Inc., 1326 Freeport Road, Pittsburgh, PA 15238.)

HOIST MANUFACTURERS INSTITUTE (HMI)

HMI 100 - Electric Wire Rope Hoists.

(Application for copies should be addressed to the Hoist Manufacturers Institute, 1326 Freeport Road, Pittsburgh, PA 15238.)

INSULATED POWER CABLE ENGINEERS ASSOCIATION (IPCEA)

S-19-81 - Rubber-Insulated Wire and Cable for the Transmission and Distribution of Electrical Energy.

(Application for copies should be addressed to the Insulated Power Cable Engineers Association, 192 Washington Street, Belmont, MA 02178.)

NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

ICS - Industrial Controls and Systems.
MG1 - Motors and Generators.

(Application for copies should be addressed to the National Electrical Manufacturers Association, 155 East 44th Street, New York, NY 10017.)

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NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

No. 70 - National Electrical Code.

(Application for copies should be addressed to the National Fire Protection Association, 470 Atlantic Avenue, Boston, MA 02210.)

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC.

National Motor Freight Classification.

(Application for copies should be addressed to the National Motor Freight Traffic Association, 1616 P Street, N.W., Washington, DC 20036.)

SOCIETY OF AUTOMOTIVE ENGINEERS, INC. (SAE)

J534 - Standard for Lubrication Fittings.

(Application for copies should be addressed to the Society of Automotive Engineers, Inc., 400 Commonwealth Drive, Warrendale, PA 15096.)

UNIFORM CLASSIFICATION COMMITTEE

Uniform Freight Classification Rules.

(Application for copies should be addressed to the Uniform Classification Committee, Tariff Publishing Officer, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.

3. REQUIREMENTS

3.1 Description. The OET cranes furnished under this specification shall conform to CMAA No. 70 or 74, except as specified herein. Each crane service class in this specification corresponds to the service class of the same designation in CMAA No. 70 or 74. The cranes shall be of the single or double girder type, indoor or outdoor type, cab or pendant controlled, and shall have the spans, capacities, speeds, and lifts all as specified (see 6.2.1). Cranes shall be new and unused and shall be furnished complete and ready for use when installed at the site designated.

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* 3.2 Standard commercial product. The crane shall, as a minimum, be in accordance with the requirements of this specification and shall be the manufacturer's standard commercial product with any added features needed to comply with the requirements. Additional or better features which are not specifically prohibited by this specification, but which are a part of the manufacturer's standard commercial product shall be included in the crane being furnished. Standard commercial product is a product which has been or will be sold on the commercial market through advertisements or manufacturer's catalogs, or brochures, and represents the latest production model(s).

* 3.3 Material. All material shall be new and unused. All material shall be as specified herein. Material not specified shall be of the same quality used for the intended purpose in commercial practice. Materials used shall be free from defects which would adversely affect the performance or maintainability of individual component or the overall assembly. All nuts, bolts, and screws shall have standard screw threads in accordance with NBS H28.

3.3.1 Structural steel. All structural steel used, as a minimum, shall conform to ASTM A36, equivalent SAE grades, or the steel manufacturer's equivalent grade, meeting the requirements of ASTM A36. Allowable stresses used in CMAA No. 70 or 74 and herein are based on ASTM A36 steels. Other materials may be used provided parts are proportioned to give comparable design factors.

3.3.2 Cast steel. Cast steel shall conform, as a minimum, to ASTM A27, grade 60-30, or A148.

3.3.3 Forged steel. Forged steel shall conform, as a minimum, to ASTM A235, A236, or A237, class to be consistent with the design tensile and yield strength used.

3.3.4 Nodular cast iron. Nodular cast iron shall conform to ASTM A439.

3.4 Service classes. Unless otherwise specified (see 6.2.1), all classes of cranes shall be designed to operate in normal ambient temperatures of 0° Fahrenheit (F) to 100°F, and in atmospheric conditions free from excessive dust, moisture, and corrosive fumes.

3.4.1 Class A1. Class A1 crane is used in installations where precise handling of valuable machinery and equipment, at slow speeds with long idle periods, is required. Capacity loads may be handled for initial installation of machinery and for infrequent maintenance.

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3.4.2 Class A2. Class A2 crane is used in installations where loads are relatively light, the speeds are slow, and a low degree of control accuracy is required. The loads may vary anywhere from no load to full capacity, with a frequency of a few lifts per day or month.

3.4.3 Class B. Class B crane is used in installations where service requirements are light, and the speeds are slow. Loads may vary from no load to full capacity, averaging 50 percent of capacity, with 2 to 5 lifts per hour. Not over 50 percent of the lifts will be at full capacity.

3.4.4 Class C. Class C crane is used in installations where service requirements are medium. Loads may vary anwhere from no load to full capacity, averaging 50 percent of full capacity, with 5 to 10 lifts per hour. Not over 50 percent of the loads will be at full capacity.

* 3.4.5 Class D - heavy service. This service covers cranes, usually cab operated, such as are used in heavy machine shops, foundries, fabricating plants, steel warehouses, and standard duty bucket and magnet operation where heavy-duty production is required, but with no specific cycle of operation. Load approaching 50 percent of the rated capacity will be handled constantly during the working period. High speeds are desirable for this type of service with 10 to 20 lifts per hour averaging 15 feet.

* 3.4.6 Class E - severe service. This type of service requires a heavy-duty crane capable of handling the rated load continuously, at high speed, in repetition throughout a stated period per day, in a predetermined cycle of operation. Applications include magnet, bucket, magnet-bucket combinations of cranes for scrap yards, cement mills, lumber-mills, and fertilizer plants, with 20 or more lifts per hour, all at rated capacity.

3.5 Rated capacity. Capacity rating of the cranes shall be in tons, of 2,000 pounds (lb) and shall be indicated on the capacity plate attached to the crane. The capacity shall be as specified (see 6.2.1). All handling accessories, such as slings and spreader bars, shall be considered as part of the load to be lifted, but the hook and the block shall not.

* 3.6 Safety. Each crane furnished shall conform to the applicable mandatory and advisory safety requirements of ANSI B30.2, B30.16 and/or HMI 100 except as modified herein.

3.7 Repair parts service. Crane manufacturer shall guarantee the availability of spare parts, for the crane furnished, for a period of not less than 10 years. Such spare parts shall be readily available when ordered by part number taken from the parts book supplied with the crane.

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* 3.8 Performance. The service class of the crane shall be based on the frequency of its intended use with various loads up to and including rated loads. The service class shall be defined identically with CMAA No. 70 or 74. The crane shall be specifically designed to perform all functions specified. No electrical part shall sustain a temperature rise exceeding the allowable limits of NEMA MG1 and ICS for the hoist motors, the trolley motors, bridge motors or controls. Measured steady running speeds shall not vary from specified values by more than 10 percent. Response to controls shall be smooth and uniform.

3.9 Structural design. Structural design shall conform to CMAA No. 70 or 74 except as specified herein.

3.9.1 Bridge girders. Bridge girders for class A1, A2, and B cranes with spans of 60 feet or more, and class C cranes with spans of 50 feet or more shall be of the welded box girder type with diaphrams, fabricated of plates, or plates and structural shapes, with longitudinal stiffeners as required. Girders for class A and B cranes under 60-foot span and class C cranes under 50-foot span shall be box girder type as specified above, plain or reinforced wide flange or American Standard beams, or special torsion box girders fabricated from either plate of structural shapes or a combination thereof.

3.9.2 Wind load. Unless otherwise specified (see 6.2.1), wind load for outdoor cranes shall be considered as 5 pounds per square foot (psf).

3.9.3 Allowable stress range. The allowable stress range for repeated loads shall not exceed the following allowable values for all class cranes:

Stress category	ASR (ksi)
A	32
B	25
C	21
D	17
E	12
F	14
G	12

3.9.4 Footwalks. When specified (see 6.2.1), a footwalk, extending completely across the crane on the bridge drive side, shall be provided on all cranes where the distance from the runway rail to the overhead obstruction is 36 inches or more. Minimum headroom from footwalk to overhead obstruction shall be not less than 48 inches. On cab controlled

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cranes, an additional footwalk, equal in length to twice the trolley length, shall be provided on the side opposite the bridge drive at the same end as the cab. Footwalks shall be antislip steel plate.

3.9.5 Operator's cab. Unless otherwise specified (see 6.2.1), cab controlled cranes shall have the cab located at one end of the crane bridge, under the bridge drive side footwalk. The cab entrance shall mate with a building mounted, crane access platform where indicated in the project specification. Unless otherwise specified (see 6.2.1), indoor cab controlled cranes shall have open type cabs. Unless otherwise specified (see 6.2.1), all cabs shall be equipped with a comfortable operator's seat and a loud gong or horn type warning device, operated by a momentary contact pushbutton.

3.9.6 Track beam. Track beams shall conform to the requirements of AISC.

3.10 Mechanical design. Mechanical design shall conform to CMAA No. 70 or 74, except as specified herein.

3.10.1 Load block and hook. Load block frames shall be of steel construction. Hook shall be of forged steel with a safety latch. Hook shall be supported on a ball or roller thrust bearing and shall rotate freely.

3.10.2 Hoisting ropes. Hoisting ropes shall be improved plow steel conforming to RR-W-410, type I, class 2 or 3, with fiber or steel core.

3.10.3 Sheaves. Sheaves shall be steel or cast iron. Pitch diameter of all running sheaves shall be not less than 24 times the rope diameter when 6 by 37 rope is furnished, or 30 times the rope diameter when 6 by 19 rope is furnished. Pitch diameter of all idler sheaves shall be not less than 1/2 times the diameter of running sheaves. Specified sheave pitch diameters shall apply to normal and low headroom cranes and cranes having special lift and clearance requirements.

3.10.4 Drum. The drum shall be steel, and shall be designed to withstand combined crushing and bending loads. Drum shall be so designed that not less than two complete wraps of the hoisting rope will remain in the grooves when the hook is at the lowest position, and not require overlapping of the rope when the hook is at the highest position. Drum grooves shall be machined. Grooving shall be right and left hand. Minimum drum groove depth shall be 3/8 times the rope diameter. Minimum drum groove pitch shall be 1.14 times the rope diameter, or the rope diameter +1/8 inch, whichever is smaller. Pitch diameter of all drums shall be not less than 24 times the rope diameter, when 6 by 37 rope is furnished,

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or 30 times the rope diameter when 6 by 19 rope is furnished. Specified drum requirements shall apply to normal and low headroom cranes and cranes having special lift and clearance requirements.

3.10.5 Gearing. All gears and pinions shall be in accordance with applicable AGMA standard.

3.10.6 Bearings. All bearings shall be of the antifriction type, except that bearings subject to a slight rocker motion, such as the idler sheave bearing, may be of the bronze bushing type. The B-10 life rating of antifriction bearings shall be 3,000 hours for class A1, A2, and B cranes, and 7,500 hours for class C cranes.

3.10.7 Brakes.

3.10.7.1 Hoist holding brakes. Hoist holding brakes shall have means for adjustment to compensate for wear.

3.10.7.2 Trolley brakes. A trolley brake shall be provided in each trolley drive.

3.10.8 Bridge drives. Unless otherwise specified (see 6.2.1), bridge drive arrangement shall be as selected by the manufacturer and shall conform to one of the arrangements in CMAA No. 70 or 74.

3.10.9 Trolley drives. Unless otherwise specified (see 6.2.1), trolley drive shall be gear type as selected by the manufacturer.

3.10.10 Couplings. All couplings shall be located immediately adjacent to a bearing. Couplings between closely spaced bearings shall be of full flexible type. Couplings for use with floating shafts shall be of the half flexible type with pilots. Rigid couplings shall be of steel or malleable iron and shall be of the compression or safety flange type.

* 3.10.11 Wheels. Maximum allowable wheel load in lb (considering dead load plus hook load only) for trolley and bridge travel wheels shall be in accordance with the following equation:

$$\text{Maximum allowable wheel load} = K \times W \times D$$

Where

K = 1600 for class A and B cranes.
  = 1400 for class C and D cranes.
  = 1200 for class E cranes.
W = Effective width of railhead in inches (width of top of head minus head corner radii).
D = Wheel tread diameter in inches.

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Trolley and bridge travel wheels shall be double flanged and of rolled-to-shape wrought or forged steel, except that wheels for pin-and-keeper axles may be of cast steel. All travel wheels shall have hardened treads, ground or machined true to diameter and width. Cylindrical driving wheel treads shall be matched within 0.001 inch per inch of diameter, but not exceeding 0.010 inch. Wheels for class A, B, and C cranes shall be rim toughened to between 320 and 370 Brinell hardness numbers. For class D and E cranes, wheel treads shall be deep carburized to not less than 550 Brinell hardness number with a gradual change of hardness with depth. Wheels on rotating axles shall be pressed on, and driving wheels shall be keyed to the axle. Hub ends of wheels on nonrotating axles shall be ground and polished to mate with bronze thrust washers at truck side members. All idler wheels and trolley drive wheels shall have cylindrical treads. Bridge drive wheels shall have treads tapered either 1 inch in 20 inches or 1 inch in 25 inches. Fillet radii between tread and flanges of wheels shall be 1/16 inch less than corner radius of rail head. Outside diameter of wheel shall be from 1-1/2 inches to 2-1/4 inches greater than tread diameter. Nominal flange thickness shall be not less than 7/8 inch. Trolley wheel treads shall be equal to the width of the railhead +1/4 inch for 8 inch and smaller wheels and +1/2 inch for larger wheels. Tread widths for bridge wheels shall be equal to the width of the railhead +1 inch for 16 inch and smaller wheels and +1-1/2 inches for wheels larger than 16 inches in diameter.

3.10.12 Bumpers. All cranes, except as otherwise permitted herein, shall be equipped with spring type bumpers. Class A1, A2, and B cranes of 5 ton or less capacity, 40 feet or less span, and travel speed of 50 foot per minute (fpm) or less may be equipped with rubber bumpers. Height of bridge bumpers above the rail shall be as specified (see 6.2.1).

* 3.10.13 Stops. All cranes shall be provided with stops in accordance with the provisions of ANSI B30.2.

3.11 Electrical equipment. The electrical equipment shall conform to CMAA No. 70 and 74, except as specified herein, and shall be designed to operate from the power supply, as specified (see 6.2.1).

* 3.11.1 Motors. Each crane motion (bridge, trolley, and hoist) shall be driven by a separate motor (or motors, for duplex bridge drive) independent of other motions. Bridge motor shall be suitable for the service class specified for the crane (see 1.2). Bridge motors shall conform to NEMA MG1 and the following:

(a) Motor mounting, shaft; and key dimensions shall conform to the manufacturer's standard.
(b) Temperature rise shall be used in accordance with NEMA MG1, for the motor type and insulation used.

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(c) Motor shall have minimum 30-minute time ratings and NEMA minimum class B insulation, except that hoist motors for speed control class 4 on class C cranes shall have minimum 60-minute time rating and NEMA class B, F, or H insulation.

(d) Squirrel-cage hoist motors shall be NEMA design D.

(e) Squirrel-cage travel drive motors shall be NEMA design A or B.

(f) Motors shall be totally enclosed and shall have ball or roller bearings.

(g) Two speed squirrel-cage motors shall have either 4:1 or 3:1 speed ratios.

3.11.2 Controls. Controls shall be full-magnetic type. There shall be a separate controller for each crane motion. The minimum size of the magnetic contractors shall be in accordance with the applicable tables contained in CMAA No. 70 or 74. When full voltage, severe inching, or plugging service is furnished, the contractors shall be rated in accordance with NEMA standards for full voltage starting and reversing of poly-phase motors. Bridge and trolley controls shall be speed control class 1 or 2, as specified (see 6.2.1). Hoist controls shall be speed control class 1, 3, or 4, as specified (see 6.2.1). When specified (see 6.2.1), hook shall be capable of lowering and lifting rated load with an adjustable speed range between 1 fpm and 15 fpm.

3.11.2.1 Speed control class 1. Speed control class 1 shall provide reversing two-speed control of a two-speed squirrel-cage motor. Speed control class 1 hoists shall have an automatic mechanical load brake. Closed transition reduced voltage starting and acceleration shall be provided for motors in 15 horsepower (hp) or greater, or for duplex-drive motors of 7.5 hp each, or greater. When specified (see 6.2.1), electronic, silicon controlled rectifier (SCR) type controls, or primary resistor reduced voltage starting, acceleration, and deceleration shall be provided for bridge and trolley service regardless of the above motor hp limit.

3.11.2.2 Speed control class 2. Speed control class 2 shall provide reversing, plugging, 5-step, secondary-resistance control of a wound rotor motor for bridge and trolley travel service only. The first running point shall provide not more than 50 percent of rated torque for cab-controlled cranes and not more than 30 percent of rated torque for floor-controlled cranes.

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3.11.2.3 Speed control class 3. Speed control class 3 shall provide reversing, 5-step, secondary-resistance control of a wound rotor hoist motor, with an automatic mechanical load-lowering brake for hoist service only. The first speed point for hoisting shall provide not more than 45 percent of full-load motor torque. The first speed point for lowering shall cause 50 percent of rated load to lower at not more than 20 percent of full-load hoisting speed. The second speed point for lowering shall cause 75 percent of rated load to lower at not more than 35 percent of full-load hoisting speed.

3.11.2.4 Speed control class 4. Speed control class 4 shall provide reversing, 5-step, secondary-resistance control of a wound rotor hoist motor, with an eddy-current load brake for hoist service only. Eddy-current braking shall be provided on four lowering, and at least two hoisting speed points. Provisions shall be made to prevent operation on any eddy-current brake controlled point upon loss of eddy-current brake excitation. The eddy-current brake shall be excited with reduced voltage when the crane's main contactor is closed and the hoist master switch is in the OFF position. Speed points shall be automatically timed and sequenced. There shall be a positive drive-down with the control set on any lowering point. The speed on the first point hoisting shall not exceed 20 percent of rated hoist speed, with no load on the hook, and shall be more than zero with 125 percent of the rated full load. The speed on the second point hoisting shall not exceed 60 percent of rated hoist speed, with no hook load, or 20 percent with full rated load. The speed on the first point lowering shall not exceed 15 percent of rated hoist speed under full load. On the second point lowering, the speed shall be between 10 percent and 20 percent of rated hoist speed, with no load, and between 30 percent and 40 percent with full load. The last or fifth point lowering shall be a regenerative point with full-load speed not more than 115 percent or rated hoist speed. The third and fourth point speeds shall be evenly spaced between the second and fifth point speeds. The crane torque rating of the eddy-current brake shall be not less than 1.25 times the 30-minute torque rating of the motor. At 100 percent of rated eddy-current brake excitation and 15 percent of rated motor speed, the braking torque developed by the eddy-current brake shall be not less than rated motor torque multiplied by the square of the hoist gear efficiency. At one-eighth of rated motor speed and with 100 percent excitation, the eddy-current brake shall develop a torque of not less than 80 percent of rated motor torque. An alternator shall be direct connected to the hoist power train. When the crane main contactor is open, the direct current output terminals of a rectifier fed by the alternator shall be connected to the eddy-current brake. Any downward movement of the hook load due to holding brake failure or slippage shall result in eddy-current brake excitation such that full-load cannot descend at greater than 50 percent speed.

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3.11.3 Enclosures. Enclosures for control panels, controls, and brakes shall be as specified (see 6.2.1).

* 3.11.4 Resistors. Resistors, when furnished, shall be in accordance with the applicable class ratings as specified in NEMA ICS.

3.11.5 Protection. Protection shall be in accordance with CMAA No. 70 or 74, and NFPA No. 70.

3.11.6 Heat shields. When specified (see 6.2.1), heat shields shall be provided to protect electrical components from high radiant heat.

3.11.7 Hoist limit switches. Upper and lower hoist limit switches shall be provided in one of the following manners: (1) a geared control circuit type upper and lower hoist limit switch plus a block-actuated control or power circuit upper hoist limit switch for safety backup; or (2) a geared control circuit lower hoist limit switch and a block-actuated power circuit upper hoist limit switch.

3.11.8 Bridge electrification system. Each bridge electrification system shall be of the enclosed safety or festooned type.

3.11.8.1 Enclosed safety type. Enclosed conductors shall be rigid shapes of galvanized steel, copper clad steel, or copper. Supports shall be spaced not greater than 4 feet apart. All supports, brackets, insulators, fastening devices, conductors, and power takeoffs necessary for complete installation shall be furnished. Plastic covers shall be of the snap-on or slip-on type, completely covering all live conductors and so shaped that a workman's finger cannot penetrate the collector openings far enough to contact live conductors. Collectors shall be double-wheel, or sliding shoe type and shall match their respective electrification systems. Collector arms shall be insulated, spring-loaded, and shall permit sufficient lateral and vertical movement to allow for imperfect track and for misalinement of the collector support in relation to the conductors.

3.11.8.2 Festooned type. Festooned type conductors shall be rope-stranded, flexible, insulated, portable cable, conforming to IPCEA S-19-81, type G. Cables shall be supported in spiral loops by swiveled wheeled carriers running in a messenger track. The carriers and messenger track shall form an integrated system manufacturerd specifically for the use intended, and shall function in a manner which protects the conductor cables from abrasion, kinking, or excessive twisting. Cables shall be coiled against the lay of cable stranding so that the cable relaxes as it is intended.

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3.11.9 Pendant pushbutton stations. Pendant pushbuttons for floor controlled cranes shall be legibly and permanently marked in the following order in a single vertical line from top to bottom: RESET; BRIDGE FORWARD, BRIDGE REVERSE; TROLLEY FORWARD, TROLLEY REVERSE; UP, DOWN (for one hoist only); AUXILIARY UP, AUXILIARY DOWN; MAIN UP, MAIN DOWN; STOP.

3.12 Welding. Welding procedures of mechanical components shall conform to the requirements of AWS D1.1 except as otherwise specified herein.

* 3.13 Cleaning, treatment, and painting. Surfaces normally painted in good commercial practice shall be cleaned, treated, and painted as specified herein. Surfaces to be painted shall be cleaned and dried to insure that they are free from contaminants such as oil, grease, welding slag and spatter, loose mill scale, water, dirt, corrosion products, or any other interfering substances. As soon as practicable, after cleaning, and before any corrosion product or other coating interfering material can result, the surface shall be prepared or treated to insure the adhesion of the coating system. The painting shall consist of at least one coat of primer and one finish coat. The primer shall be applied to a clean, dry surface as soon as practicable after cleaning and treating. Painting shall be with manufacturer's current materials according to manufacturer's current processes and the total dry film thickness shall be not less than 2.5 mils over the entire surface. The paint shall be free from runs, sags, orange peel, or other defects. The color of the finish coat shall be as specified (see 6.2.1).

* 3.14 Dissimilar metals. Intimate contact which can be expected to cause galvanic corrosion shall be avoided. When such contact cannot be avoided, an interposing insulating material shall be provided to minimize the corrosive effect.

* 3.15 Lubrication. Lubrication shall be in accordance with the equipment manufacturer's recommendations and shall be compatible with lubricants specified herein. All parts requiring lubrication shall be lubricated prior to delivery and tagged to show the type and temperature rating of the lubricant used.

* 3.16.1 Lubrication fittings. Hydraulic lubrication fittings shall be in accordance with SAE J534. Pressure lubrication fittings shall not be used where normal lubricating pressure may damage grease seals or other parts.

* 3.16 Fungus resistance. When specified (see 6.2.1), electrical components and circuit elements, including terminal and circuit connections, shall be coated with varnish conforming to MIL-V-173, except that:

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(a) Components and elements inherently inert to fungi or in hermetically sealed enclosures need not be coated.
(b) Current-carrying contact surfaces, such as relay contact points, shall not be coated.

* 3.17 Electromagnetic interference characteristics. When specified (see 6.2.1), the equipment shall conform to the electromagnetic characterists requirements and test limits for class No. IIB equipment as specified in MIL-STD-461.

* 3.18 Identification plates. Unless otherwise specified (see 6.2.1), identification plates shall be furnished by the manufacturer. They shall conform to MIL-P-514, type I, style 1, composition C. Plates shall be affixed to the crane in a conspicuous place with brass screws or bolts of not less than 1/8-inch diameter.

* 3.19 Erection. When specified (see 6.2.1), the contractor shall provide erection services to completely erect, adjust, lubricate, and make the crane ready for operation at the erection site.

* 3.20 Capacity plates. Two capacity plates shall be provided on each crane, one for each side of the bridge. Each plate shall be lettered to indicate the rated hoisting capacity of each hoist in lb, with letters of sufficient size to be easily read from the floor. Beneath the rated capacity, or capacities, there shall be placed in smaller letters, the name of the procuring agency (such as Naval Facilities Engineering Command), the contract number under which the crane is purchased, and the manufacturer's serial or crane number. Name and capacity plates shall be of heavy-sheet steel with porcelain enamel face of black background and white letters or white background with black letters.

* 3.21 Instruction plates and wiring diagrams. Where applicable, the equipment shall be provided with instruction plates describing special or important procedures for operating and servicing equipment, and warnings or cautions of hazardous procedures. The plates shall be durable and shall remain legible throughout the life of the equipment. Complete wiring diagrams, including schematics, shall be mounted behind control panel covers under clear plastic.

* 3.22 Workmanship. Workmanship shall be in accordance with the best standard practice of the crane industry.

* 3.22.1 Steel fabrication. Steel used in the fabrication of equipment shall be free from kinks and sharp bends. The straightening of material shall be done by methods that will not cause injury to the metal. Shearing and chipping shall be done neatly and accurately. Corners shall be square and true. Flame cutting [illegible]

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for the thickness of metal, may be employed instead of shearing or sawing. All bends of a major character shall be made with metal dies or fixtures in order to insure uniformity of size and shape. Precautions shall be taken to avoid overheating, and heated metal shall be allowed to cool slowly.

* 3.22.2 Bolted connections and fasteners. Bolt holes shall be accurately punched or drilled and shall have the burrs removed. All bolts, nuts, and screws shall be tight. All fastenings or parts which might become loosened by vibration, shock, or operation of the crane shall be secured by suitable locking devices.

* 3.22.3 Welders and welding operators. Before assigning any welder or welding operator to work covered by this specification, the contractor shall provide the contracting officer with the names of welders and welding operators to be employed in the work, together with certification that each of these welders and welding operators has passed the qualification tests as prescribed by AWS D1.1 for the type of welding operation to be performed, and that such qualification is effective as defined by the particular code. The contractor shall require any welder or welding operator to retake the tests when, in the opinion of the Government inspector, the work of the welder or welding operator creates a reasonable doubt as to his proficiency. Tests, when required, shall be conducted at no additonal expense to the Government. Recertification of welders or welding operators shall be made to the contracting officer only after the welders or welding operators have taken and passed the required retests.

* 3.22.4 Castings. All castings shall be sound and free from patching, misplaced coring, warping, or any other defect which reduces the castings ability to perform its intended function.

* 3.22.5 Machine work. Metal fits, tolerances, and finishes shall conform to those indicated on the approved drawings. Finished contact and bearing surfaces shall be true and exact.

* 3.23 Disassembly for shipment. Crane disassembly for shipment shall be the minimum practicable. The bridge girders shall be shipped full length with as much of the bridge drive and footwalks assembled as physically practicable. The end trucks and trolley(s) shall be shipped completely assembled. The cab shall be shipped as a complete unit. Parts which are disassembled for shipment shall be matchmarked and tagged with metal tags. The tags and marking thereon shall be waterproofed.

* 3.24 Technical data. When specified (see 6.2.1), drawings and design data shall be forwarded in accordance with the contract data requirements included in the contract.

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* 3.24.1 Design evaluation. The design evaluation drawings shall provide all essential information needed to permit an evaluation of the proposed design (see 6.3). The drawings shall include, but not be limited to, the following:

* 3.24.1.1 Arrangement drawings. The arrangement drawings shall show all main features of the system, including track layout, clearances, lifts, speeds, maximum crane wheel loads (without impact), required air distribution systems, electrical conductor layout, and connections with the electrical power source, as applicable.

* 3.24.1.2 Design and calculation drawings. The design calculations shall include material designations, calculations, design loads, stresses, safety factors, and deflections as applicable for cranes, track beams, electrification, and supports as applicable.

* 3.24.1.3 Electrical drawings. Schematic and complete wiring diagrams shall be furnished, identifying all electrical equipment including manufacturer's name, model number, wire types, and sizes as applicable.

* 3.24.2 Technical manuals. Complete operating and maintenance instructions and parts list shall be forwarded in accordance with the contract (see 6.3).

* 3.25 Servicing and restoration. Each unit tested shall be serviced and restored to a service condition equal to the original condition of the unit, neglecting nominal wear incurred during the tests. The restoration shall include paint touchup or repainting, as required for delivery.

3.26 Shop assembly. When specified (see 6.2.1), each crane shall be assembled, except for the rope reeving, at the contractor's plant, wired, and examined and tested as specified (see 4.2 and 4.3). Cabs shall be completely assembled with all glazing, equipment, conduit, wiring, and other parts. For test purposes, completely assembled cabs need not be mounted on the bridge or trolley and may be electrically connected to the crane with temporary wiring. Hydraulic lines for brakes need not be assembled in the shop.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.2 Examination. Each crane shall be examined for compliance with the requirements specified in Section 3 of this specification. Any redesign or modification of the contractor's standard product to comply with specified requirements, or any necessary redesign or modification following failure to meet specified requirements, shall receive particular attention for adequacy and suitability. This element of inspection shall encompass all visual and tactile examinations and dimensional measurements. Noncompliance with any specified requirement or presence of one or more defects preventing or lessening maximum efficiency shall constitute cause for rejection.

4.3 Shop tests. Each crane, after being assembled as specified (see 3.26) at the contractor's plant, shall be tested without load to determine successful operation of the assembled unit. The assembled crane shall be square and in alinement, with parts fitted and adjusted properly. Operation shall be smooth and quiet. Bearings shall be cool running. Motor controls, as applicable, shall be timed and adjusted to give smooth acceleration and minimum current peaks. After completion of motor control adjustments, the protective equipment shall be set to function at slightly higher than normal peak current and slightly lower than normal low voltage.

4.4 Operating tests. The bridge and trolley brakes shall be adjusted by the contractor so as to stop the bridge and trolley without excessive swinging of the hoist ropes. After the completed cranes are erected in place, adjusted, lubricated, and otherwise made ready for operation, they will be tested by the Government, in the presence of the contractor, as indicated on Test Forms A through H inclusive, which are attached to this specification. Bridge and trolley travel and hoisting tests for speed will be made after mechanisms have attained essentially steady state, prior to taking data. The Government will furnish the labor, power, slings, and weights required for the operating tests. Tests will be conducted observing the safety rules of applicable ANSI standards.

4.5 Electromagnetic interference characteristics tests. When specified (see 6.2.1), the equipment shall be tested in accordance with MIL-STD-462 to determine conformance to the electromagnetic interference characteristics required by MIL-STD-461. Unless otherwise specified, tests shall be conducted by the contractor on the complete crane at the erection site.

4.6 Packaging inspection. The preservation-packaging, packing, and marking of the cranes shall be inspected to verify conformance to the requirements of Section 5.

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5. PACKAGING

5.1 Preservation-packaging, and packing. Unless otherwise specified (see 6.2.1), the complete crane shall be preserved and packaged in accordance with the manufacturer's standard practice, and shall be packed in a manner which will insure arrival in satisfactory condition, and which will be acceptable to the carrier at lowest rates. Containers and packing shall comply with Uniform Freight Classification rules or National Motor Freight Classification rules. Cranes for overseas destinations shall be preserved, packaged, and packed as specified (see 6.2.1).

6. NOTES

6.1 Intended use. OET cranes are used in shop, maintenance, assembly, repair, and storage areas. OET cranes covered by this specification are intended to be procured as a part of a construction project covered by a project specification. Allowable options (when specified, unless otherwise specified, and as specified) in this specification will be furnished as detailed in the project specification.

* 6.2 Ordering data. Procurement documents should specify:

6.2.1 Procurement requirements.

(a) Title, number, and date of this specification.
(b) Type and service class of crane required (see 1.2).
(c) Whether crane is to be indoor or outdoor type, cab or pendant controlled, and the spans, capacities, hoist speeds, trolley travel speeds, bridge travel speeds, and lifts required (see 3.1).
(d) When temperature and atmosphere conditions are other than as specified (see 3.4).
(e) Capacity of crane required (see 3.5).
(f) Wind load, in psf, if different from that specified in 3.9.2.
(g) When footwalks are required (see 3.9.4).
(h) When cab controlled cranes shall have cab located different from that specified in 3.9.5.
(i) When indoor cab controlled cranes shall have cab different from that specified in 3.9.5.
(j) When operator's seat and warning device are not required in cab controlled cranes (see 3.9.5).
(k) When bridge drive arrangement shall be different from that specified (see 3.10.8).
(l) Type of gear drives required (see 3.10.9).
(m) Height of bridge bumpers (see 3.10.12).
(n) Power supply characteristics (see 3.11).

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(o) Speed control class for bridge, trolley, and hoist controls (see 3.11.2).
(p) When hook lowering and lifting speed shall be as specified (see 3.11.2).
(q) When reduced voltage starting, acceleration, and deceleration required for speed control class 1 trolley and bridge travel service with squirrel-cage motors less than the limit specified in 3.11.2.1.
(r) Type of enclosures for control panels, controls, and brakes (see 3.11.3).
(s) When heat shields are required (see 3.11.6).
(t) Color of finish coat is required (see 3.13).
(u) When fungus control is required (see 3.16).
(v) When electromagnetic interference testing is required (see 3.17).
(w) When identification plate is required (see 3.18).
(x) When erection service is required (see 3.19).
(y) When technical data is required (see 3.24).
(z) When shop assembly and tests are required (see 3.26 and 4.3).
(aa) When electromagnetic interference characteristics tests are required (see 4.5).
(bb) When preservation-packaging, and packing is different from that specified in 5.1.

* 6.3 Contract data requirements. When this specification is used in a procurement which incorporates a DD Form 1423 and invokes the provisions of paragraph 7-104.9(n) of the Armed Services Procurement Regulation (ASPR), the data requirements identified below will be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the Contract Data Requirements List (DD Form 1423) incorporated into the contract. When the provisions of ASPR 7-104.9(n) are not invoked, the data specified below shall be delivered in accordance with the contract requirements. Deliverable data required by this specification is cited in the following paragraphs:

Paragraph	Data requirements	Applicable DD 1664
3.24.1	Drawings, Design Evaluation	UDI-E-24001A
3.24.2	Manuals, Technical Modified Commercial	UDI-M-24006C

(Copies of Data Item Descriptions required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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* 6.4 Schematic drawing. The procurement documents should include a schematic drawing showing the following information.

(a) Plan of the building and runways, giving length of runways, spans of runway girders, location and details of runway bumper stops, building access platforms and gates, mezzanines, obstructions, size and locations of pits to be serviced by the hooks, location of crane cab or pendant control station (horizontally and vertically), location of crane in relation to other cranes, cardinal directions, and if critical, the hook approaches towards the ends of the building.

(b) Cross section of the building and runways, showing crane and giving span, runway rail type and size, height of top of rail from floor, maximum hook heights required, lowest drop of the hook required, end approaches of the hooks toward the side of the building, clear distance from center of runway rails to columns or other obstructions, such as conduit or ducts, clear distance, or distances from the top of the runway rails to overhead building construction or other obstructions, such as light fixtures or ducts, required clearances at these points, height and dimensions of access platform on building structure; size and type, location, and spacing of runway conductors; and location of crane cab or pendant control station.

(c) Cross section of the runway girder showing member sizes, web stiffeners, runway rail and method of securing it to the runway girders.

(d) The wheel loads (without impact) and wheel spacing which were used to design the runway; bumper heights and sizes of bumping faces of existing cranes on the same runway; kind of electric current, its voltage, phase, and frequency; and if other cranes are on the runway, the maximum (or minimum, if applicable) distance between hooks of the new crane and the adjacent existing crane, with existing crane data as necessary for the contractor to design his end trucks, walkways, and cab.

* 6.5 Changes from previous issue. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, correction, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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Navy - YD

Preparing activity:

Navy - YD

Project No. 3950-N167

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FORM A

Test of (Main) (Auxiliary) Hoist

No Load on Hook

	Up					Down				
	1st	2nd	3rd	4th	5th	1st	2nd	3rd	4th	5th
Points										
Volts										
Amperes										
Motor, rpm										
Block speed, fpm										
Accleration										
Controller timing										

Alinement

Lubrication

Vibration

Brake(s)

Temperature of bearings ____, motor ____, resistors ____, reactors ____.

Upper limit switch setting ____, and drift ____.

Lower limit switch setting ____, and drift ____.

Remarks:

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FORM B

Test of (Main) (Auxiliary) Hoist

25 Percent of Rated Capacity on Hook

Points	Up					Down				
	1st	2nd	3rd	4th	5th	1st	2nd	3rd	4th	5th
Volts										
Amperes										
Motor, rpm										
Block speed, fpm										
Accleration										
Controller timing										

Alinement

Lubrication

Vibration

Brake(s)

Temperature of bearings ____, motor ____, resistors ____, reactors ____.

Upper limit switch setting ____, and drift ____.

Lower limit switch setting ____, and drift ____.

Remarks:

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FORM C

Test of (Main) (Auxiliary) Hoist

50 Percent of Rated Capacity on Hook

	Up					Down				
	1st	2nd	3rd	4th	5th	1st	2nd	3rd	4th	5th
Points										
Volts										
Amperes										
Motor, rpm										
Block speed, fpm										
Accleration										
Controller timing										

Alinement

Lubrication

Vibration

Brake(s)

Temperature of bearings ____, motor ____, resistors ____, reactors ____.

Upper limit switch setting ____, and drift ____.

Lower limit switch setting ____, and drift ____.

Remarks:

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FORM D

Test of (Main) (Auxiliary) Hoist

75 Percent of Rated Capacity on Hook

Points	Up					Down				
	1st	2nd	3rd	4th	5th	1st	2nd	3rd	4th	5th
Volts										
Amperes										
Motor, rpm										
Block speed, fpm										
Accleration										
Controller timing										

Alinement

Lubrication

Vibration

Brake(s)

Temperature of bearings ____, motor ____, resistors ____, reactors ____.

Upper limit switch setting ____, and drift ____.

Lower limit switch setting ____, and drift ____.

Remarks:

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FORM E

Test of (Main) (Auxiliary) Hoist

100 Percent of Rated Capacity on Hook

	Up					Down				
	1st	2nd	3rd	4th	5th	1st	2nd	3rd	4th	5th
Points										
Volts										
Amperes										
Motor, rpm										
Block speed, fpm										
Accleration										
Controller timing										

Alinement

Lubrication

Vibration

Brake(s)

Temperature of bearings ____, motor ____, resistors ____, reactors ____.

Upper limit switch setting ____, and drift ____.

Lower limit switch setting ____, and drift ____.

Remarks:

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FORM F

Trolley Travel Test

Rated Capacity on Hook

	Up					Down				
	1st	2nd	3rd	4th	5th	1st	2nd	3rd	4th	5th
Points										
Volts										
Amperes										
Motor, rpm										
Block speed, fpm										
Accleration										
Controller timing										

Alinement

Lubrication

Vibration

Brake(s)

Temperature of bearings ____, motor ____, resistors ____, reactors ____.

Upper limit switch setting ____, and drift ____.

Lower limit switch setting ____, and drift ____.

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FORM G

Bridge Travel Test

Rated Capacity on Hook, Centered Between Runways

	Up					Down				
	1st	2nd	3rd	4th	5th	1st	2nd	3rd	4th	5th
Points										
Volts										
Amperes										
Motor, rpm										
Block speed, fpm										
Accleration										
Controller timing										
Alinement										
Lubrication										
Vibration										
Brake(s)										

Temperature of bearings ____, motor ____, resistors ____, reactors ____.

Upper limit switch setting ____, and drift ____.

Lower limit switch setting ____, and drift ____.

Remarks:

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FORM H

Test of (Main) (Auxiliary) Hoist

125 Percent of Rated Capacity on Hook

	Up					Down				
Points	1st	2nd	3rd	4th	5th	1st	2nd	3rd	4th	5th
Volts										
Amperes										
Motor, rpm										
Block speed, fpm										
Accleration										
Controller timing										

Alinement

Lubrication

Vibration

Brake(s)

Temperature of bearings ____, motor ____, resistors ____, reactors ____.

Upper limit switch setting ____, and drift ____.

Lower limit switch setting ____, and drift ____.

Remarks:

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NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*:
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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